Outcomes, Events, and Probability

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1 Introduction

To "compile" this information I am using the book **A modern Introduction to Probability and Statistics** by F.M. Dekking, C. Kraaikamp, H.P. Lopuhaä, L.E. Mcester. It can be found here: `https://link.springer.com/book/10.1007/1-84628-168-7`.

This document is designed to review the Probability and Statistic principles covered in the aforementioned book. I will create other "Guides" for additional topics. They can be found at: `https://github.com/KovasMcCann/Writings`.

2 Sample Spaces

A **sample space** is the set of all possible outcomes of an experiment. Each element in the sample space represents a potential result of the experiment we are interested in.

For example, the sample space for a single coin toss is:

$$\Omega = \{H, T\}$$

where H represents heads and T represents tails.

In general, the sample space includes all possible outcomes that can occur in a given experiment.

3 Events

An **event** is a subset of the sample space. It consists of one or more outcomes from the sample space that share a certain property of interest.

4 Probability

A **probability** is how likley and event is to occur.

The Formal Definition:

A probability function P on a finite sample space Ω assigns to each event A in Ω a number $P(A)$ in $[0, 1]$ such that (i) $P(\Omega) = 1$, and (ii) $P(A \cup B) = P(A) + P(B)$ if A and B are disjoint. The number $P(A)$ is called the probability that A occurs.

For example in a coin flip the probability is:

$$P(H) = P(T) = \frac{1}{2}$$

Where $P(H)$ is the probability of getting heads and $P(T)$ is the probability of getting tales.

5 Addition Rule

The **addition rule** helps us find the probability that at least one of two events occurs.

- **For independent events:**

$$P(A \cup B) = P(A) + P(B)$$

- **General events:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The subtraction of $P(A \cap B)$ ensures that outcomes common to both A and B are not counted twice.

6 Multiplication Rule

The **Multiplication rule** helps us find the probability that both of two events occur.

- **For independent events:**

$$P(A \cap B) = P(A) \cdot P(B)$$

- **General events:**

$$P(A \cap B) = P(A) \cdot P(B|A)$$

The conditional probability $P(B|A)$ is the probability of B given that A has occurred.

7 Complement Rule

The **complement rule** states that the probability of an event not occurring is equal to one minus the probability of the event occurring.

If A is an event, then the complement of A, denoted A^c, is the event that A does not occur. The complement rule is given by:

$$P(A^c) = 1 - P(A)$$

8 Products of Sample Spaces

When considering multiple experiments, the sample space of the combined experiment is formed by the **Cartesian product** of the individual sample spaces.

- **Independent events:** If there are n independent experiments, each with the same sample space S, then the total number of possible outcomes is

$$|S|^n$$

where $|S|$ is the number of outcomes in S.

- **General case:** If the experiments have sample spaces $S_1, S_2, \ldots, S_n$, then the combined sample space is

$$S_1 \times S_2 \times \cdots \times S_n$$

and the total number of possible outcomes is

$$|S_1| \times |S_2| \times \cdots \times |S_n|$$

For dependent events, the sample space is still the Cartesian product, but probabilities may depend on previous outcomes (conditional probabilities)

9 Infinite Sample Spaces

An **infinite sample space** is one that contains an infinite number of outcomes. This can occur in various scenarios, such as when dealing with continuous variables or unbounded processes. For example, consider the sample space of all possible outcomes when rolling a fair die infinitely many times. The sample space can be represented as:

$$\Omega = \{(x_1, x_2, x_3, \ldots) \mid x_i \in \{1, 2, 3, 4, 5, 6\} \text{ for } i = 1, 2, 3, \ldots\}$$

In this case, each outcome is an infinite sequence of die rolls, where each x_i represents the result of the i-th roll. The sample space is uncountably infinite because there are infinitely many possible sequences of outcomes.